START HERE

Test your part on the computer before you build it.

Paramfabric shows you how a part bends, shakes, heats up, or how air and water flow around it. You load a 3D model, say where it is held and pushed, press **Run Simulation**, and read the colours.

This manual assumes you have never run a simulation. Every step names the exact button you press.

THE WHOLE IDEA IN FOUR STEPS

1. **Geometry.** Open your 3D part (or the demo bracket).
2. **Physics.** Say what you want to know: will it bend (FEA) or how does air flow (CFD). Mark where it is held and pushed.
3. **Mesh & solve.** The computer cuts the part into thousands of tiny pieces and does the maths. One button.
4. **Results.** Blue is calm, red is stressed or fast. Check the numbers against the rules in this manual.

01 Should I use Paramfabric for this?

DECISION HELPER

Answer three questions. The helper tells you which mode to use, or when to use something else.

1 · WHAT DO YOU WANT TO FIND OUT?

- **Will it bend or break?** (selected)
- Will it vibrate or resonate?
- How does air or water flow?
- Will it overheat?
- Crash, explosion or impact
- Keep dimensions in sync

2 · WHAT IS THE PART LIKE?

- **A chunky solid part** (selected)
- Thin sheets or frames
- Rubber, plastic that yields, or very large bending

3 · WHAT IS IT FOR?

- **Comparing designs early** (selected)
- Evidence for a review or audit
- Certifying something lives depend on

RECOMMENDATION

Use FEA · linear static

GOOD FIT

Physics → FEA. Mark a Fixed support and a Force. Run. Check stress and displacement in section 05.

02 Before you start

SET-UP

Open the Studio

On Windows, double-click **Start Paramfabric.bat**. The Studio opens in its own window. If you use a terminal: `python -m paramfabric.desktop`.

Airflow needs OpenFOAM

Bending, vibration and beams work out of the box. For airflow and heat, install OpenFOAM (on Windows, inside WSL Ubuntu). The Studio finds it by itself; the solver list says *OpenFOAM* when it is ready.

Use STEP files

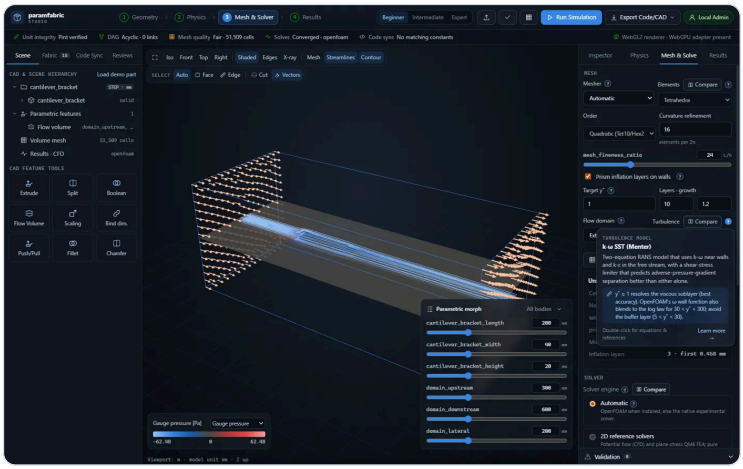
Export your part from your CAD program as **STEP** (.step / .stp). STL works for viewing, but meshing and editing need STEP or IGES.

Pick your guidance level in the top bar. *Beginner* shows a `?` badge on every setting with a plain explanation and safe defaults. *Expert* hides the help and unlocks advanced solver controls. Start with Beginner.

03 The four steps, screen by screen

STEP 1 OF 4

Geometry & Parameters

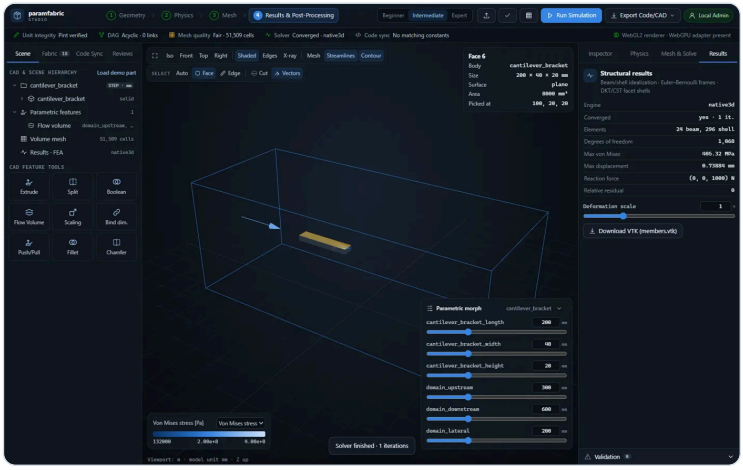


This is where your part lives. The left side lists your parts and their dimensions; the middle is the 3D view.

1. Click **Import CAD** and choose a STEP file, or click **Load demo part**.
2. Drag to rotate, right-drag to pan, scroll to zoom. **Iso**, **Front**, **Top** snap the view.
3. In the **Parametric morph** box, drag a slider. The real CAD shape changes, not just the picture.
4. Open the **Fabric** tab to see every dimension with its unit and the reason it was chosen.

STEP 2 OF 4

Physics & Boundary Conditions

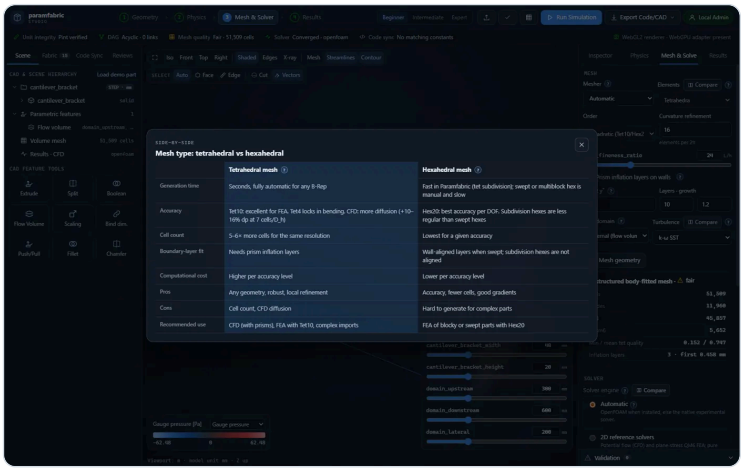


Tell the computer what physics to use and where the part is held and pushed. This is the step that decides whether the answer is right.

1. Open the **Physics** tab on the right. Pick **FEA** (strength) or **CFD** (flow).
2. Press **Create default physics parameters**. Check the material values (steel is 200 GPa).
3. Click a face in the 3D view. In the Inspector press **Fixed support**, **Force** or **Pressure**.
4. No supports set? The Studio clamps the left end and pushes down on the right end, and says so.

STEP 3 OF 4

Mesh & Solver



The computer splits the part into small pieces (the mesh) and solves equations on each one. Smaller pieces are more accurate and slower.

1. Open **Mesh & Solve**. Leave the mesh on its default the first time.
2. Press **Mesh Geometry** to see the pieces, or go straight to **Run Simulation**.
3. While it runs, the residual graph should fall. A flat or rising line means trouble (see Help).
4. To check accuracy, make the mesh finer and run again. If the answer barely moves, you are done.

STEP 4 OF 4

Results & Post-Processing

Colours show the answer: blue is low, light or red is high. The legend at the bottom-left gives the numbers.

1. Change the field in the legend: stress or displacement for FEA, pressure or velocity for CFD.
2. Click **Cut** and drag the slider to look inside. **Vectors** shows flow arrows.
3. Click **Iso-surface** to show every place where a value equals your chosen level.
4. Press **Export Compliance Audit (PDF)** to save a signed report of everything.

04 Three tutorials

HANDS-ON

TUTORIAL A · 10 MINUTES

Will this shelf bracket bend too much?

1. Click **Load demo part**. You get a 200 × 40 × 20 mm steel bracket.
2. Open **Physics**, choose **FEA · linear static**, press **Create default physics parameters**.
3. Click the left end face of the bracket, then **Fixed support**. Click the right end face, then **Force**.
4. Press **Run Simulation** (`Ctrl` + `Enter`).
5. In **Results**, read *Max displacement* and *Max von Mises*. Put them in the checker in section 05.

Expect roughly 0.5 mm and 75 MPa for 1 kN. Steel S355 yields at 355 MPa, so this bracket is safe by a factor of about 4.7.

TUTORIAL B · 15 MINUTES

How does air flow around it?

1. With the demo part loaded, open **Physics** and choose **CFD · external flow**.
2. Press **Extract flow volume**. A box of air appears around the part.
3. Press **Run Simulation**. Watch the residual graph fall; that means the answer is settling.
4. Click **Cut** in the viewport toolbar and drag the slider to slice through the air. Turn on **Vectors**.
5. Click **Iso-surface**, choose *Velocity magnitude*, drag the slider low to see the slow wake.

Read *Drag coefficient* and *Wall y^+* in Results. If y^+ looks odd, see Help.

TUTORIAL C · 10 MINUTES

A frame made of beams and sheets

1. In FEA, click **Edge** in the Select toolbar, then click an edge of the part.
2. The **Assign 1D beam** window opens. Pick *I-beam*, type the sizes, watch the drawing, press **Assign**.
3. Click **Face**, click a flat face. In **Assign 2D shell** set the thickness, press **Assign**.
4. Press **Run Simulation**. Beams show as coloured lines, sheets as coloured surfaces.

Use beams and shells when the walls are thin (thickness less than 1/20 of the span). They solve in seconds.

Will it vibrate? (modal analysis)

In **Physics**, set *Structural analysis* to **Modal** and choose how many modes. Run. Results lists each natural frequency in Hz; click a row to see that shape. If a motor, fan or road runs near one of those frequencies, the part will shake. Keep at least 20% away.

Will it overheat? (heat transfer)

In CFD, tick **Conjugate heat transfer**, choose the coolant and the solid, enter the heater power in watts. Select the face that heats up and press **Heat source**. Run. Results gives the hottest solid temperature and checks that the heat in equals the heat out.

05 Is my result OK?

RESULT CHECKER

Colours look convincing even when they are wrong. Put your numbers here and get a plain answer.

Material

Structural steel S355 · yield 355 MPa

Max von Mises stress from Results **75 MPa**

Safety factor you need **2×**

Span or length of the part **200 mm**

Max displacement from Results **0.5 mm**

Allowed deflection rule

Span / 250 (general structures)

STRENGTH

Safe · factor 4.73

Yield 355 MPa is 4.7× the peak stress; you asked for 2×.

STIFFNESS

Stiff enough

Allowed: 200 mm / 250 = 0.80 mm. Yours: 0.50 mm.

```
safety factor = yield / stress = 355 / 75 = 4.73
· deflection limit = span / 250 = 0.80 mm
```

Also check: the peak is not just one red dot at a sharp corner (sharp inside corners show fake high stress; add a fillet), the load and supports are on the right faces, and refining the mesh does not change the answer by more than a few percent.

06 Words you will see

GLOSSARY

CAD

The 3D drawing of your part. Paramfabric reads STEP and IGES files.

Parameter

A named number with a unit, like bracket_length = 200 mm. Change it once; everything updates.

Fabric

The list of all parameters, their formulas and why they were chosen.

Unit check

Paramfabric refuses to add millimetres to newtons, and converts inches to millimetres for you.

FEA

Finite element analysis: works out how a solid part bends and how stressed it is.

CFD

Computational fluid dynamics: works out how air or water moves and pushes on the part.

Mesh

The part cut into many small pieces (elements or cells). The maths is done on each piece.

Boundary condition

What happens at the edges: where it is held (support), pushed (force), or where air enters (inlet).

Fixed support

A face that cannot move at all, like a bolted end.

von Mises stress

One number for how hard the material is working at a point. Compare it with the yield strength.

Yield strength

The stress at which a metal bends permanently. Stay well below it.

Safety factor

Yield strength divided by your peak stress. 2 means it could take twice the load before yielding.

Displacement

How far a point moves under load, in millimetres or metres.

Natural frequency

The rate (Hz) at which a part likes to vibrate. Keep motors and fans away from it.

Mode shape

The pattern a part vibrates in at one natural frequency.

Beam (1D)

A long thin member drawn as a line, with a cross-section like an I-beam or tube.

Shell (2D)

A thin wall drawn as a surface, with a thickness.

Residual

How far the solver is from a settled answer. It should fall steadily during a run.

Converged

The answer has stopped changing. Results show yes or no.

Turbulence model

A shortcut for chaotic airflow. k-ω SST is the safe default.

y^+ (y-plus)

How fine the mesh is right next to a wall in CFD. The Studio reports it so you can check it.

Named selection

A group of faces, edges or bodies you gave a name. It keeps pointing at the same geometry when dimensions change.

Cutting plane

A slice through the part or the air so you can see inside.

Iso-surface

A surface joining every point where a value is the same, like a contour line in 3D.

Streamline

The path a particle of air would follow through the flow.

Heat source

A face that puts heat into the part, in watts.

Compliance report

A signed PDF listing every parameter, check and result, for reviews and audits.

Signature

A digital seal. If anyone changes the report, verification fails.

07 When something goes wrong

“Import CAD geometry before meshing” –

Load a part first: Import CAD or Load demo part.

“Faceted model; unstructured meshing needs STEP/IGES” –

You opened an STL. Export the part from your CAD as STEP and import that.

“Missing physics parameter” –

Open Physics and press Create default physics parameters.

The whole part is red –

The load may be far too large, or a unit is wrong (N vs kN, mm vs m). Check the Fabric tab; units are shown next to every value.

The residual line goes flat or up –

The flow is not settling. Refine the mesh near the part, lower the relaxation in Expert mode, or add iterations.

Result says “not reliable” –

You ran CFD without OpenFOAM. Install OpenFOAM and run again; the solver list shows when it is found.

One tiny red spot at a corner –

Sharp inside corners give unreal peak stress. Add a fillet, or read the stress a little away from the corner.

Heat balance error above 2% –

Run more iterations. The answer is not settled until heat in equals heat out.

Verify a report says “edited after signing” –

The file was changed after export. Ask for the original, or export a fresh one.

Golden rules

- A simulation answers the question you asked. Wrong supports give a wrong answer with beautiful colours.
- Start coarse, then refine. If the answer moves less than 5% when you refine the mesh, trust it more.
- Compare with a hand calculation once. Sections 05 and the demo numbers are there for that.
- Results marked `not reliable` are for exploring only.
- For anything people's safety depends on, a qualified engineer must check the result.

Prove your report to someone

Press **Export Compliance Audit (PDF)**. Send the PDF and the file from **Public key**. They press **Verify a report...** or run `paramfabric report-verify report.pdf --public-key key.pem`. If anyone edits the PDF, it fails.

08 Right-click anything

The menu only shows what makes sense for the thing you clicked. If you are not sure what you can do with a face or an edge, right-click it.

Right-clicking a face in FEA. Hold and drag with the right button to pan, as before.

RIGHT-CLICK ON	YOU CAN
A face	Look at it head-on, select similar faces, name it, add a support or load, assign a shell, hide its body
An edge	Look at it, name it, assign a beam, fillet or chamfer
A result colour	Pin the exact value there, draw an iso-surface at that value, cut through that point
Empty space	Change the view, go back to the previous view, change the display, save a picture
A row in the Scene list	Rename, zoom to it, delete a feature
A parameter	Edit it, copy its name or value

Name the faces that matter

1. Right-click a face and choose **Create named selection...**. Type a name such as *Bolted end*. Hold `Ctrl` and click first to include several faces.
2. It appears under **Named selections** in the Scene list. Click it to find those faces again.
3. Right-click it and choose **Insert boundary condition**. Now the support stays on that face even after you change the part's dimensions.

Keyboard shortcuts

`L`	Look at the selection	`Z`	Zoom to it
`H`	Hide its body	`I`	Show only its body
`A`	Show all bodies	`N`	Name the selection
`F2`	Rename	`F`	Fit everything
`Esc`	Clear the selection	`Ctrl` + click	Add to the selection

Hiding only changes what you see. Hidden bodies are still part of the simulation.

09 Make the Studio yours

WINDOWS, MOUSE, MESH

Move things around

- Drag the thin line between panels to resize them. Double-click it to hide a panel.
- Drag a floating window by its title. Double-click the title to put it back.
- Your layout is remembered. Right-click empty space and choose **Reset window layout** to start over.
- `Ctrl` + `=` makes everything larger, `Ctrl` + `0` resets it.

Use the mouse you know

Right-click empty space and open **Mouse controls**. Choose SolidWorks, Ansys Mechanical, Siemens NX or Autodesk Fusion, and dragging works the way you are used to.

The coloured axes at the top right show which way you are looking. Click one to look along it.

Cannot find something?

Press `Ctrl` + `P` and type what you want, such as *mesh*, *section* or *report*. Press `?` for every shortcut.

`Ctrl` + `Z` undoes changes to settings and parameters.

Is my mesh any good?

1. After **Mesh geometry**, read the sentence under **Mesh viewer**. It tells you in plain words.
2. The bars show cell shapes: left is bad, right is ideal. Most cells should sit on the right. Click a bar to see those cells in red.
3. Press **Go to worst cell**. If it sits right where you care about the answer, refine the mesh there.
4. Turn on **Cut** to see the real cells inside the part.

When a run fails

A red card stays on screen and tells you what happened and what to try. Open **Technical details** for the raw output, and use **Copy details** if you need to ask someone for help.

While a run is going, the graph header shows its speed and the longest it can still take. It often finishes sooner.

Paramfabric Studio 0.6 · Academy. Material yield values are typical minimums; use your supplier's certified values. Deflection limits are common rules of thumb; your code or customer may set different ones.